



main.c...

Output



```
1  #include <iostream>
2  #include <iomanip>
3  #include <limits>
4  using namespace std;
5  const int LOW = 60, HIGH = 100;
6  int main() {
7      // -----
8      // Task 4.1: Heart-Rate Monitor
9      // (do-while)
10     // -----
11     cout << "input: 72 55 110 80 -1"
12         << endl << endl;
13     int test_bpms[] = {72, 55, 110, 80
14         , -1};
15     int count = 0;
16     double sum = 0;
17     double mn = numeric_limits<double>
18         >::max();
19     double mx = numeric_limits<double>
20         >::lowest();
21     int i = 0;
22     int bpm;
23     do {
24         bpm = test_bpms[i++];
25         if (bpm >= 0) {
26             count++;
27         }
28     } while (bpm != -1);
29 }
```

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22     sum += bpm;
23     if (bpm < mn) mn = bpm;
24     if (bpm > mx) mx = bpm;
25     if (bpm < LOW) {
        cout << "*** BRADYCARDIA: " <<
        bpm << " BPM ***" << endl;
26     } else if (bpm > HIGH) {
27         cout << "*** TACHYCARDIA: " <<
        bpm << " BPM ***" << endl;
28     }
29     }
30     } while (bpm >= 0);
31     if (count > 0) {
32         double avg = sum / count;
33         cout << "Count:" << count << "
            Avg:" << avg << " Min:" << mn
            << " Max:" << mx << endl;
34     }
35     // -----
        -----
36     // Task 4.2: Dosage Schedule Table
        (for)
37     // -----
        -----
38     cout << "\n4.2" << endl;
39     cout << "output:" << endl << endl;
40     cout << "    " << setw(4) << "
        << "    " << "CumDose(mg,"

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```
40      cout << "      " << setw(4) << "Hour"
      << "      " << "CumDose(mg)" <<
      endl;
41      int dose = 5, cum = 0;
42      for (int hour = 0; hour <= 24;
      hour += 6) {
43          cum += dose;
44          if (hour == 0 || hour == 6 ||
      hour == 24) {
45              cout << "          " << hour <<
      "          " << cum <<
      endl;
46          }
47      }
48      // -----
      -----
49      // Task 4.3: Image Patch
      Intensities (for)
50      // -----
      -----
51      cout << "\n4.3" << endl;
52      cout << "Input: 10" << endl <<
      endl;
53      int N = 10;
54      for (int index = 0; index < N;
      index++) {
55          int intensity = index % 256;
56          cout << "Pixel " << index << "
      : " << intensity << endl;
```

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```
43     hour += 6) {
44     cum += dose;
45     if (hour == 0 || hour == 6 ||
46         hour == 24) {
47         cout << "      " << hour <<
48             "      " << cum <<
49             endl;
50     }
51 }
52 // -----
53 // Task 4.3: Image Patch
54 // Intensities (for)
55 // -----
56 cout << "\n4.3" << endl;
57 cout << "Input: 10" << endl <<
58     endl;
59 int N = 10;
60 for (int index = 0; index < N;
61     index++) {
62     int intensity = index % 256;
63     cout << "Pixel " << index << "
64         : " << intensity << endl;
65 }
66 return 0;
67 }
```

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input: 72 55 110 80 -1

*** BRADYCARDIA: 55 BPM ***

*** TACHYCARDIA: 110 BPM ***

Count:4 Avg:79.25 Min:55 Max:110

4.2

output:

Hour	CumDose(mg)
0	5
6	10
24	25

4.3

Input: 10

Pixel 0: 0

Pixel 1: 1

Pixel 2: 2

Pixel 3: 3

Pixel 4: 4

Pixel 5: 5

Pixel 6: 6

Pixel 7: 7

Pixel 8: 8

Pixel 9: 9



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```
1  #include <iostream>
2  #include <iomanip>
3  #include <limits>
4  using namespace std;
5  int main() {
6      cout << "ECG stream (then -999): ";
7      int ecg_stream[] = {40, 78, 150,
8                          60, 35, 140, 55, 160, 48, 30,
9                          130, -999};
10     int sample_count = 0;
11     int high_samples = 0;
12     int sum = 0;
13     int mn = numeric_limits<int>::max
14         ();
15     int mx = numeric_limits<int>
16         >::lowest();
17     const int THRESHOLD = 100;
18     int i = 0;
19     int sample;
20     while (true) {
21         sample = ecg_stream[i++];
22         cout << sample << " ";
23         if (sample == -999) {
24             break;
25         }
26         sample_count++;
27         sum += sample;
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```
25     if (sample < mn) mn = sample;
26     if (sample > mx) mx = sample;
27     if (sample > THRESHOLD) {
28         high_samples++;
29     }
30 }
31 cout << endl << endl;
32 cout << "--- REPORT ---" << endl;
33 if (sample_count > 0) {
34     double avg = (double)sum /
35                 sample_count;
36     int range = mx - mn;
37     // Heart rate estimation
38     formula: (High Samples *
39              60000) / (Sample Count * 4
40              )
41     double bpm = ((double)high_samples *
42                  60000.0) / (sample_count * 4.0);
43     cout << "Number of samples: "
44           << sample_count << endl;
45     cout << "Mean (average) sample
46           value: " << fixed <<
47           setprecision(2) << avg <<
48           endl;
49     cout << "Minimum sample value:
50           " << mn << endl;
51     cout << "Maximum sample value: "
52           << mx << endl;
```

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```
        setprecision(2) << avg <<
        endl;
40      cout << "Minimum sample value:
        " << mn << endl;
41      cout << "Maximum sample value:
        " << mx << endl;
42      cout << "Range (max - min): "
        << range << endl;
43      cout << "Samples exceeding
        THRESHOLD (100): " <<
        high_samples << endl;
44      cout << "Estimated Heart Rate:
        " << setprecision(2) <<
        bpm << " BPM" << endl;
45      cout << "Classification: ";
46      if (bpm < 60) {
47          cout << "BRADY" << endl;
48      } else if (bpm >= 60 && bpm <=
        100) {
49          cout << "NORMAL" << endl;
50      } else {
51          cout << "TACHY" << endl;
52      }
53  }
54
55      return 0;
56  }
57  |
```

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Output



```
ECG stream (then -999): 40 78 150 60 35 140
                        55 160 48 30 130 -999
```

```
--- REPORT ---
```

```
Number of samples: 11
```

```
Mean (average) sample value: 84.18
```

```
Minimum sample value: 30
```

```
Maximum sample value: 160
```

```
Range (max - min): 130
```

```
Samples exceeding THRESHOLD (100): 4
```

```
Estimated Heart Rate: 5454.55 BPM
```

```
Classification: TACHY
```

```
=== Code Execution Successful ===
```



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github.com



Shahzaib
ShahzaibRafiqueKhan



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